

5b. Sensitivity Analysis

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1/ Sensitivity Analysis

Cf. Cinelli & Hazlett (2020), *Making Sense of Sensitivity*, JRSS-B

Sensitivity Analysis for Regression

- Standard regression estimator of the ATE:

$$Y_i = \hat{\alpha} + \hat{\tau}D_i + \mathbf{X}'_i\hat{\beta} + \varepsilon_i$$

- Note: applies equally to fully interacted models (cf. W4).

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- Standard **omitted variable bias (OVB)** formula:

$$\begin{aligned}\hat{\tau} &= \tau + \gamma \times \underbrace{\frac{\text{cov}(D_i^{\perp \mathbf{X}}, U_i^{\perp \mathbf{X}})}{\mathbb{V}(D_i^{\perp \mathbf{X}})}}_{\text{regression of } U_i^{\perp \mathbf{X}} \text{ on } D_i^{\perp \mathbf{X}}} \\ &= \tau + \underbrace{\gamma \times \delta}_{\text{OVB}}\end{aligned}$$

- Gives us $\rightsquigarrow \tau = \hat{\tau} - \gamma\delta$

Partial R-Squared Interpretations

- Standard OVB written in terms of regression coefficients is difficult to reason about.
- Easier to reason with partial R^2 version of OVB (Cinelli and Hazlett, JRSSB 2020).
 - Intuition: bias $\propto$ how much U predicts $Y \times$ how much U predicts D .

$$|\text{bias}| = \sqrt{\frac{R^2_{Y \sim U|D, \mathbf{x}} \cdot R^2_{D \sim U|\mathbf{x}}}{1 - R^2_{D \sim U|\mathbf{x}}}} \cdot \frac{\mathbb{V}(Y \perp\!\!\!\perp \mathbf{x}, D)}{\mathbb{V}(D \perp\!\!\!\perp \mathbf{x})}$$

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- Partial R^2 is the incremental predictive value of one variable:

$$R^2_{Y \sim U|D, \mathbf{x}} = \frac{R^2_{Y \sim D + \mathbf{x} + U} - R^2_{Y \sim D + \mathbf{x}}}{1 - R^2_{Y \sim D + \mathbf{x}}} = \frac{\text{additional variance explained by } U}{\text{variance unexplained by } D, \mathbf{x}}$$

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- Partial R^2 is the incremental predictive value of one variable:

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- **Sensitivity analysis** can then vary two unknown parameters:
 - $R^2_{Y \sim U|D, \mathbf{x}} \in [0, 1]$: incremental predictive value of U for the outcome.
 - $R^2_{D \sim U|\mathbf{x}} \in [0, 1]$: incremental predictive value of U for treatment.
 - From these we can determine the bias and thus the true value of τ .

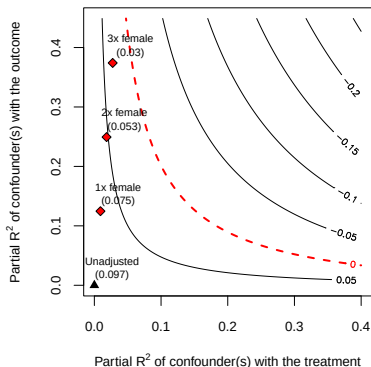
R Package: sensemakr

```
1 # load package
2 library(sensemakr)
3
4 # import dataset
5 data("darfur")
6
7 # run the regression model
8 model <- lm(peacefactor ~ directlyharmed + age + farmer_dar + herder_dar +
9             pastvoted + hhsize_darfur + female + village, data = darfur)
10
11 # conduct the sensitivity analysis
12 sensitivity <- sensemakr(model = model,
13                           treatment = "directlyharmed",
14                           benchmark_covariates = "female",
15                           kd = 1:3)
```

- Darfur conflict (Sudan, 2003) survey: D = directly harmed by violence, Y = support for peacemaking.
- female is the strongest known confounder $\rightsquigarrow$ used as benchmark.
 - kd (knock-on distance): how many times stronger would the unobserved confounder need to be (relative to female) to change our conclusion?

Sensitivity Analysis with Partial R^2

```
1 # plot bias contour of point estimate  
2 plot(sensitivity)
```



- Contour lines = adjusted estimate $\hat{\tau}$.
- Red diamonds = benchmarks at 1–3× strength of female.

- Here: even a confounder 3× as strong as female would not eliminate the effect. $\rightsquigarrow$ Our estimate is robust to plausible levels of unobserved confounding.